A-A-53417
28 March 1988

COMMERCIAL ITEM DESCRIPTION

BUR, BONE SURGERY (Swanson Reamer)

The General Services Administration has authorized the use of this Commercial Item Description.

Salient Characteristics:

This commercial item description covers Swanson Reamer surgical buffing burs of the following head diameters:

2mm
3mm
4mm

Material. The burs shall be fabricated from Class 5, Type 630, corrosion-resistant steel in accordance with ASTM F899.

Hardness. The burs shall be properly hardened and passivated. After proper stress relieving, the burs shall have a Rockwell hardness of C40 to C44, when tested as specified herein.

Style and dimensions. The burs shall be of one-piece construction and shall be furnished in the head diameters specified in Figure 1.

The burs shall be of Swanson Reamer design suitable for surgical buffing.

Burs shall be compatible for use with medium bur guard, component of PNEUMATIC BONE DRILL (See notes).

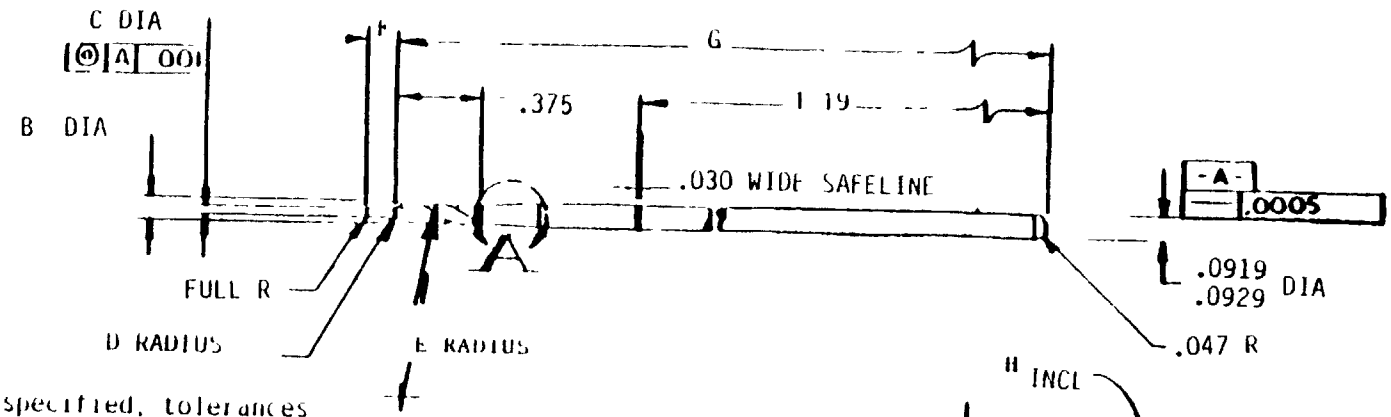
The proximal end of the burs shall be round and non-notched.

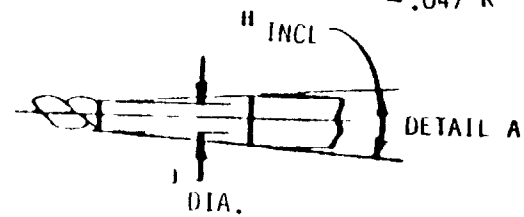
The operative end cutting edges of the burs shall be uniformly cut, shaped and sharpened. The burs shall have a non-cutting tip and shall conform to figure 1 for design and dimensions for the applicable NSN and head diameter (See notes).

AMSC N/A FSC 6515

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Unless otherwise specified, tolerances are.

.XX ± .03
.XXX ± 010
Angles ± 0° 30'

	HEAD DIAMETER (mm)	B	C	D	E	F	G	H	J
6515-01-139-7738	2	078 ± 005	031 ± 005	062	1 250	125	2.04	$1^{0} \pm 1^{0}$	.062 / .058
6515-01-139-7739	3	118 ± 005	062 ± 005	125	1 00	156	2.27	$4^{0} \pm 1^{0}$	072 / .068
6515-01-139-7740	4	157 ± 005	062 ± 005	.125	875	.156	2.27	$4^{0} \pm 1^{0}$	.072 / .068

NOTES

1 Head shape: RH Cutting, Non-cutting tip (blunt), 4 teeth (2mm), 6 teeth (3mm and 4mm), 30^{0} RH Spiral

2 Finish #32

3 Remove all burrs

4. Unless otherwise specified dimensions are in inches

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FIGURE 1 BUR, BONE SURGERY, Swanson Beam

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Concentricity and straightness. The burs shall meet the concentricity and straightness requirements specified in figure 1.

Runout. The total indicated runout of the burs shall not exceed 0.001 inch when measured as specified herein.

Marking. The burs shall be marked with a safeline as specified in Figure 1.

Finish. The finish of the burs shall be as specified in figure 1. The operative end shall have a fine honed finish with sharp cutting edges. The safeline shall have an etched finish.

Corrosion resistance. The burs shall not show evidence of corrosion when tested as specified herein.

Workmanship. The burs shall have uniformly cut and formed teeth and shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One bur, of the stock number specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of II. Unit of product for examination purposes shall be one bur.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as specified below. Unit of product for test purposes shall be one bur.

Test Criteria.

Characteristic	Inspection level	AQL (Percent defective)
Chemical analysis	*	**
Dimensions	II	1.5
Hardness	S-1	**
Runout	S-1	1.0
Corrosion resistance	S-1	1.0

* Not applicable.

** Acceptance number shall be zero for all sample sizes.

Sampling of raw materials. A sufficient quantity of each lot of raw steel used in the fabrication of the instruments shall be selected for chemical analysis.

Examination. The burs shall be examined to determine compliance with the requirements contained in this specification. Examination shall include, but shall not be limited to, the defects listed below.

Defects.

Head diameter not as specified.
Design not as specified.
Bur not of one-piece construction.
Cutting edges not sharp.
Cutting edges not uniformly cut and shaped.
Finish of bur head not as specified.

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Laminations, cracks, bends, deformed portions, gouges, chips, or broken areas present in any part of bur.
Tip not blunt.
Safeline missing, not clearly defined or not of uniform tickness.
Finish of shank not as specified.
Number of teeth not as specified for applicable bur size.

Microscopic inspection. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 7 power (7X) magnification, unless otherwise specified herein.

Tests.

Chemical analysis. In the absence of certificates of quality, the chemical analysis of the instruments shall be determined in accordance with method 111 or 112 of FED-STD-151. In the event of conflict, method 111 shall govern.

Dimensions test. Tests of dimensions shall be made with a tool maker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

Hardness. Hardness shall be determined in accordance with American Society for Testing and Materials (ASTM) Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to .0002 inch and equipped with a knife-edge foot, a shadowgraph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter nearest the head. The maximum and minimum readings obtained during one complete 360° rotation of the bur shall be measured and recorded to the nearest .0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Corrosion resistnce. The burs shall meet the requirements specified herein when subjected to the boil test as specified in ASTM 1089.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTES:

1. The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6515-01-139-7738	BUR, BONE SURGERY, Swanson Reamer, 2.0mm Head Diameter
6515-01-139-7739	BUR, BONE SURGERY, Swanson Reamer, 3.0mm Head Diameter
6515-01-139-7740	BUR, BONE SURGERY, Swanson Reamer, 4.0mm Head Diameter

2. The following National Stock Number is referenced by this document:

National Stock Number	Item Identification
6515-01-204-2664	DRILL, BONE, PNEUMATIC, Nitrogen or Compressed Air

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

Project No. 6515-4200

Location: NCR Disk #4

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-50417	BUR, BONE SURGERI (Swanson Reamer)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify) _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) - Optional	b. WORK TELEPHONE NUMBER (Include Area Code) - Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) - Optional	8. DATE OF SUBMISSION (YYMMDD)

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